

A counterexample to the all-vector law of the iterated logarithm for random weighted shifts

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21 August 2026

Result. Conjecture 7.17 of Cheng–Fang–Zhu is false. There are bounded positive i.i.d. weights satisfying all of the paper’s hypotheses and a deterministic nonzero vector $x \in \ell^2$ for which

$$\frac{\log \|T^n x\|^2}{\sqrt{2\sigma^2 n \log \log n}} \rightarrow +\infty \quad \text{almost surely.}$$

1 The source conjecture

For a unilateral random weighted shift $Te_k = X_k e_{k+1}$, Lemma 7.15 of [1] proves, for each basis vector e_k , the usual LIL with normalization $\sqrt{2\sigma^2 n \log \log n}$, where $\mathbb{E} \log X_1 = 0$ and $\sigma^2 = \mathbb{E}(\log X_1)^2 < \infty$. Conjecture 7.17 asks whether the same assertion holds for every $x \in \ell^2$.

2 Proof intuition

For a fixed basis vector, $\|T^n e_k\|$ sees only one length- n weight block, so the ordinary random-walk LIL applies. An infinite-support vector is different: its norm is a sum over all starting locations and can select a rare exceptionally favorable block far out in the tail.

With symmetric weights $e^{\pm 1}$, an all- e block has probability 2^{-n} . Among $\lceil e^n \rceil$ disjoint blocks, the probability of seeing none is double-exponentially small and hence summable. The coefficient $k^{-3/5}$ decays slowly enough that an all- e block starting before ne^n still contributes $\exp((4/5)n - O(\log n))$ to the squared norm. Borel–Cantelli therefore upgrades this rare-block mechanism to an almost-sure lower bound for every sufficiently large n , far above the $\sqrt{n \log \log n}$ LIL scale.

3 Counterexample

Theorem 1. Let $X_j = e$ and $X_j = e^{-1}$, each with probability $1/2$, independently, and set

$$x = \sum_{k=1}^{\infty} k^{-3/5} e_k.$$

Then $x \in \ell^2$, the weights satisfy the hypotheses of Lemma 7.15 with $\sigma^2 = 4$, and almost surely

$$\lim_{n \rightarrow \infty} \frac{\log \|T^n x\|^2}{\sqrt{8n \log \log n}} = +\infty.$$

In particular Conjecture 7.17 is false.

Lemma 7.15. *Let $T \sim \{X_n\}_{n=1}^\infty$, $\{e_k\}_{k=1}^\infty$ be the associated orthonormal basis for $\mathcal{H}$, $\mathbb{E}(\ln X_1) = 0$ and $\mathbb{E}(\ln X_1^2) = \sigma^2 < \infty$. Then for each k ,*

$$(47) \quad \limsup_{n \rightarrow \infty} \frac{\ln \|T^n e_k\|^2}{\sqrt{2\sigma^2 n \ln \ln n}} = 1, \quad \liminf_{n \rightarrow \infty} \frac{\ln \|T^n e_k\|^2}{\sqrt{2\sigma^2 n \ln \ln n}} = -1 \quad a.s.$$

Indeed, to be more specific, for any $\varepsilon > 0$, let

$$A_n = \left[\|T^n e_k\|^2 < e^{(1+\varepsilon)\sqrt{2\sigma^2 n \ln \ln n}} \right], \quad B_n = \left[\|T^n e_k\|^2 > e^{(-1-\varepsilon)\sqrt{2\sigma^2 n \ln \ln n}} \right].$$

Then

$$(48) \quad \mathbb{P}(\liminf_{n \rightarrow \infty} A_n) = \mathbb{P}(\liminf_{n \rightarrow \infty} B_n) = 1.$$

In order to treat the random model, one might need more estimates beyond the mere growth rate in the mean. Then the tail probability of $\|T^n e_k\|^2$ offered by, say, the classical Hoeffding's inequality [57] or the like, can be helpful.

For a general moment sequence $\{\|T^n x\|^2\}_{n=1}^\infty$, let $x = \sum_{k=1}^\infty a_k e_k \in \mathcal{H}$. Then

$$(49) \quad \|T^n x\|^2 = \sum_{k=1}^\infty |a_k|^2 X_k^2 \cdots X_{k+n-1}^2.$$

Remark 7.16. This is a sum of inter-correlated random variables from a stationary series. For sums of independent r.v.'s, plenty of techniques are available [80, 64]. Unfortunately, for dependent random variables, existing knowledge in probability does not readily yield satisfactory answers for us. For instance, one can apply the theory of orthogonal series to the above (49) if $\mathbb{E}(X_k^2) = 0$, which is absurd for us. As mentioned in the introduction, we plan to start a separate work to sort out the general theory of (8) and (9) which are elegant examples of dependent summation.

Conjecture 7.17. Lemma 7.15 holds for $x \in \mathcal{H}$ (instead of only $e_k \in \mathcal{H}$).

8. ALGEBRAS GENERATED BY T

The aim of this section is to describe various algebras generated by a random weighted shift $T(\omega)$. The list includes Banach algebras, weakly closed algebras, commutant algebras/multipliers, dual algebras and C^* -algebras. The proofs rest on some past results, especially from [88].

8.1. Non-selfadjoint algebras. Let $A \in \mathcal{B}(\mathcal{H})$. We denote

$$\mathcal{A}_A = \text{the norm closure of polynomials in } A$$

and

$$\mathcal{W}_A = \text{the closure in the weak operator topology of polynomials in } A.$$

Thus $\mathcal{A}_A$ is the Banach algebra generated by A . By [32 P. 477], $\mathcal{W}_A$ coincides with the closure, in the strong operator topology, of polynomials in A . We denote by $\{A\}'$ the commutant of A , and by $\{A\}''$ the double commutant of A . It is easy to see that $\{A\}'$ and $\{A\}''$ are both closed in the weak operator topology and $\mathcal{W}_A \subset \{A\}'$.

Since $\mathcal{B}(\mathcal{H})$ is the dual space of the ideal of trace-class operators, it makes sense to define

$$\mathcal{O}_A = \text{the closure, in weak* topology, of polynomials in } A,$$

which is called the *dual algebra* generated by A [7].

Theorem 8.1. *Let $T \sim \{X_n\}_{n=1}^\infty$ with $R = 1$. Then*

Figure 1: Lemma 7.15, formula (49), and Conjecture 7.17 in arXiv:1811.05761, PDF page 48.

Proof. Plainly $\sum k^{-6/5} < \infty$, while $\mathbb{E} \log X_1 = 0$ and $\mathbb{E}(\log X_1^2)^2 = \mathbb{E}(\pm 2)^2 = 4$.

Fix n , put $M_n = \lceil e^n \rceil$, and consider the M_n disjoint blocks of indices

$$B_{n,r} = \{1 + rn, \dots, (r+1)n\}, \quad 0 \leq r < M_n.$$

Let F_n be the event that none of these blocks consists entirely of weights equal to e . Independence within this fixed collection gives

$$\mathbb{P}(F_n) = (1 - 2^{-n})^{M_n} \leq \exp(-M_n 2^{-n}) \leq \exp(-(e/2)^n).$$

The last sequence is summable, so the first Borel–Cantelli lemma implies that, almost surely, F_n occurs for only finitely many n . Notice that no independence between different values of n is needed.

Hence, almost surely for every sufficiently large n , there is an $r < M_n$ such that all weights in $B_{n,r}$ equal e . With $k = 1 + rn \leq nM_n$, formula (49) of the source then yields the one-term lower bound

$$\|T^n x\|^2 = \sum_{j \geq 1} j^{-6/5} X_j^2 \cdots X_{j+n-1}^2 \geq k^{-6/5} e^{2n} \geq (nM_n)^{-6/5} e^{2n}.$$

Since $M_n \leq 2e^n$ for $n \geq 1$,

$$\log \|T^n x\|^2 \geq \frac{4}{5}n - \frac{6}{5}\log n - \frac{6}{5}\log 2.$$

Dividing by $\sqrt{8n \log \log n} = o(n)$ proves the assertion. $\square$

4 Scope and verification

The construction exploits infinite spatial support: at time n , the norm simultaneously samples exponentially many remote length- n blocks. It does not contradict Lemma 7.15 for any fixed basis vector. The proof uses only an exact norm identity and the first Borel–Cantelli lemma. The accompanying script checks the numerical inequalities used above for a range of n ; those checks are transcription aids, not a substitute for the analytic proof.

A bounded arXiv search through 21 August 2026 using the paper id, title, authors, and the exact conjecture found no later paper recording this counterexample. This is not an exhaustive literature review.

References

- [1] G. Cheng, X. Fang, and S. Zhu, *Random weighted shifts*, arXiv:1811.05761.